AI Use Declaration

CpE 211 Laboratory — Lab 2: Linear Search (Lab_02_Linear_Search)

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Section	BSCpE-2
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Files covered	LibraryFinder.java · README.md · AI_Use_Declaration.pdf (this document) · Viva_Question_Explanation.pdf

1. Declaration

I declare that I used a generative AI coding assistant, within the instructor's AI-use policy, in preparing this laboratory work (Lab 2: Linear Search). Assisted work spans two sessions: the creation of *LibraryFinder.java* and its companion documents (October 23, 2026), and the restructuring of *README.md* together with the two PDF documents submitted with this lab (October 24, 2026). The specific uses are disclosed in this document. All outputs produced with AI assistance were reviewed, traced, tested, and verified by me before submission, and the submitted work reflects my own understanding and effort. I take full responsibility for its content.

2. AI Tool and Model

- **Assistant:** opencode (OpenCode CLI) — a terminal AI coding agent, used in both sessions
- **Model:** GLM 5.3 Flash, model ID *glm-5.3-flash-free* (provider: Z.ai) — the same model ID recorded in the October 23, 2026 declaration
- **Instructor policy:** use of this tool is permitted for laboratory work with mandatory disclosure (this declaration)
- **Scope of assistance:** the Java source *LibraryFinder.java* and its companion documents (earlier session); the *README.md*; and the two PDF documents (this session)
- **Nature of use:** pseudocode-first tutoring with an explain-back step, implementation after requirements were locked in, documentation drafting, and document generation from student-provided content

3. Prompts Used (verbatim)

3.1 Earlier session — creation of LibraryFinder.java (Oct 23, 2026)

Stage	Prompt (verbatim)	Purpose
1. Pseudocode only	"Act as a Java tutor. Help me design pseudocode for Linear Search in Real Applications. Do not write full code until I explain the algorithm back to you."	Constrain the AI to teach pseudocode only, no implementation.

2. Explain-back + build spec	"So a linear search is basically going through each item from start to end until you find a match or run out of items. I do not know why we return -1 and not 0. Best case is the target is at the start ... Build LibraryFinder with *findFirst(String[] titles, String target)* and *countMatches(String[] titles, String target)*. Print every inspected index."	Explain-back check while asking for corrections; stated the required method signatures and trace printing.
3. Manual trace spec	"Manual trace: Trace searches for 'Java' and 'Python' in [C, Java, DSA, Java, SQL]. Count every comparison. Target at first position. Target at last position. Target missing. Repeated titles including adjacent duplicates."	Locked the exact test arrays, the four required trace cases, and comparison counting.
4. Implementation approval	"You may write now. After writing, tell me how to run the code."	Approved implementation after refining requirements.
5. Declaration request	"Make an AI-use declaration in pdf. AI-use declaration: instructor-approved model, prompts used, AI suggestion adopted/rejected, and my verification."	Requested the original October 23 declaration document.

3.2 Current session — README.md and the two PDFs (Oct 24, 2026)

Stage	Prompt / Response (verbatim)	Purpose
README update	"Modify the readme.md so that it shows a short description of the project, installation/usage, and tech stack/requirements. Only access the lab2 folder."	Requested the README restructure into description, installation/usage, and tech-stack sections.
Plan approval	"Go ahead."	Approved the proposed README structure.
PDF creation request	"Also create 2 pdfs. One pdf is dedicated to AI-use-declaration. State the model, prompts used, AI suggestion adopted/rejected, and my verification. The other is for answering the question: "Why does Linear Search work on unsorted data, and when should it be avoided?". Ask me the question before finalizing anything."	Requested the two PDFs and their mandatory declaration contents.
Filename correction	"PDF 2 filename should be Viva Question Explanation."	Corrected the naming of the second PDF.
Final instructions	"Use underscores for punctuation. For question number 2, use whatever is recommended."	Filenames use underscores; the recommended answer depth was applied.

Questionnaire responses	Declared AI-assisted scope: "Code + README + PDFs"; prompts to be stated verbatim; adopted/rejected items drafted for the student's confirmation; verification performed: "Compiled & ran the program"; identity details as shown in the header above. The student's own answer to the viva question was provided as: > *"It doesnt care about the order because it checks each element one-by-one. It does not assume any order made. Linear search should be avoided for very large datasets because it goes through every element and compares it. Lets say a list has 1 million items. That would mean the worst-case scenario would be that it makes 1 million comparisons."*	Student-supplied details used as input for the two PDFs.

Note: rows 1–5 of table 3.1 were recovered from the student's original October 23, 2026 declaration (AI-Use-Declaration.pdf), which recorded them chronologically and lightly edited for spelling; rows 6–10 are quoted as typed.

4. AI Suggestions — Adopted

4.1 Earlier session (LibraryFinder.java)

- **Pseudocode-first workflow:** the design was presented as pseudocode, with implementation allowed only after the explain-back step.
- **Sentinel -1** for "not found" in *findFirst()* (0 excluded because it is a valid array index), and 0 matches for *countMatches()*.
- **Early-return design:** *findFirst()* returns on the first match (best case O(1)) while *countMatches()* performs a full scan (always Θ(n)) so a later match would not be missed.
- **equalsIgnoreCase()** for case-insensitive title matching, with null-argument guards.
- **Embedded comparison counter:** each search prints "Comparisons made: N", so the program output doubles as verifiable test evidence.
- **Two test arrays:** [C, Java, DSA, Java, SQL] (first / last / missing / repeated cases) and [Java, Java, DSA, SQL] (adjacent duplicates).

4.2 Current session (README.md and the two PDFs)

- Restructure *README.md* into **Description**, **Installation / Usage**, and **Tech Stack / Requirements** sections, including the *javac*/*java* build-and-run commands. *(Verified by executing the documented commands.)*
- Rewrite *README.md* from UTF-16 (LE) encoding to plain UTF-8 so the file renders normally in editors and viewers.
- Document the six built-in demo scenarios in the README description: target at the first position, last position, repeated title, missing target, count on repeated title, and adjacent duplicates.

- Generate the two required PDFs by writing styled HTML pages and printing them with Microsoft Edge's headless mode.

5. AI Suggestions — Rejected / Modified

5.1 Earlier session (LibraryFinder.java)

- **Case-sensitive matching** (String.equals) was recommended as the simpler lab default — **modified:** chose case-insensitive *equalsIgnoreCase()* as more realistic for a library search box.
- A **generic six-title demo array** with two duplicated titles created by the AI — **rejected:** replaced with the trace array [C, Java, DSA, Java, SQL] plus an adjacent-duplicate array, per the manual trace specification.
- The AI was ready to **write the implementation immediately** after prompt 2 — **deferred:** switched to build mode only after the manual-trace requirements were finalized.

5.2 Current session (README.md and the two PDFs)

- Rejected the AI's offer to write a full one-page essay answer for the viva question PDF. The student's own key points (one-by-one inspection, no assumption of order, and avoidance on very large datasets with the 1,000,000-comparison example) were used instead, expanded only lightly.
- Rejected the option of leaving identity placeholders (e.g., "[Student Name]") in the documents; the student's actual details were supplied and used.

6. Student Verification

6.1 Earlier session (LibraryFinder.java)

- Explained the linear search algorithm back to the AI before any code was written; the AI corrected two misconceptions that were then incorporated: (a) why **-1** is used instead of 0 as the sentinel, and (b) that the best case is O(1) while the overall bound is O(n).
- Produced the manual trace **by hand, before running any code**, counting every comparison: *findFirst("Java")* = 2 comparisons (index 1; the duplicate at 3 never inspected), *findFirst("Python")* = 5 comparisons (missing → -1), plus the first-position, last-position, and adjacent-duplicate cases.
- Compiled (*javac LibraryFinder.java*) and ran (*java LibraryFinder*), then compared the program's "Comparisons made" values against the hand trace: all 8 searches matched exactly (1, 5, 2, 5, 5, 5, 1, 4).
- Reviewed the generated code line by line against the approved pseudocode plan and confirmed it contains the inspected-index printing, early-return logic, full-scan counting, null guards, and case-insensitive comparison as specified.
- Can rebuild and explain this solution (sentinel value, early exit vs. full scan, best/worst-case complexity) without AI assistance.

6.2 Current session (README.md and the two PDFs)

- Compiled the program with *javac LibraryFinder.java* and ran it with *java LibraryFinder*.
- Reviewed the console traces for the demo scenarios — the "Inspecting index" lines, "Comparisons made" counts, first-match indices (and "Not found (-1)" results), and "Total matches" counts — and confirmed the counts behave as expected for a linear search (e.g., a target at index 0 takes 1 comparison; a missing target in a 5-item array takes 5 comparisons).
- Read and reviewed the final *README.md* and the two PDF documents before submission.

All AI-assisted output described above was reproduced locally and verified by me through execution before this declaration was finalized.

7. Academic Honesty Statement

I attest that the AI assistance disclosed above was used within the instructor's AI-use policy, that I verified all AI output against my own hand-produced manual trace as described in Section 6, and that I take full responsibility for any inaccuracy in the submitted work. Portions of this declaration itself were drafted with AI assistance for formatting.

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10/23/2026